

```

q1.java > Java > q1 > main(String[] args)
1  public class q1{
2      private String title;
3      private String author;
4      private double price;
5      public q1(String title,String author,double price){
6          this.title = title;
7          this.author = author;
8          this.price = price;
9      }
10
11      public String getTitle() {
12          return title;
13      }
14
15      public String getAuthor() {
16          return author;
17      }
18
19      public void setTitle(String title) {
20          this.title = title;
21      }
22
23      public void setAuthor(String author) {
24          this.author = author;
25      }
26
27      public void setPrice(double price) {
28          this.price = price;
29      }
30  public static void main(String[] args) {q1 book = new q1(title: "Developing Java Software", author: "Russel Winder", price: 79.75);
31      System.out.println(book.getTitle());
32      System.out.println(book.getAuthor());
33  }
34
35  }

```

Problems Output Debug Console Terminal Ports

PS D:\Lab1> ^C

PS D:\Lab1>

PS D:\Lab1> d; cd 'd:\Lab1'; & 'C:\Program Files\Microsoft\jdk-21.0.12.8-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\Lenovo\AppData\Roaming\Developing Java Software'

Russel Winder

PS D:\Lab1>

```

J q3java > java > q3 > employee
1 public class q3 {
2
3     private Employee boss;
4     private Employee employee;
5
6     public q3(Employee boss, Employee employee) {
7         this.boss = boss;
8         this.employee = employee;
9     }
10
11    public void printEmployee() {
12        System.out.println(employee.getFirstname());
13        System.out.println(employee.getLastname());
14        System.out.println(employee.getMonthlysalary());
15    }
16
17    public void printAll() {
18        System.out.println(x: "Boss:");
19        System.out.println(boss.getFirstname());
20        System.out.println(boss.getLastname());
21        System.out.println(boss.getMonthlysalary());
22
23        System.out.println(x: "Employee:");
24        System.out.println(employee.getFirstname());
25        System.out.println(employee.getLastname());
26        System.out.println(employee.getMonthlysalary());
27    }
28
29    public void updateSalary(String firstname, double salary) {
30        if (employee.getFirstname().equals(firstname) && salary > 0) {
31            employee.setMonthlysalary(salary);
32        }
33    }
34
35    public void raiseAll() {
36        boss.setMonthlysalary(boss.getMonthlysalary() * 1.10);
37        employee.setMonthlysalary(employee.getMonthlysalary() * 1.10);
38    }
39
40    Run main | Debug main | Run | Debug
41    public static void main(String[] args) {
42
43        Employee boss = new Employee(firstname: "John", lastname: "Smith", monthlysalary: 5000);
44        Employee employee = new Employee(firstname: "Alice", lastname: "Brown", monthlysalary: 3000);
45
46        q3 team = new q3(boss, employee);
47
48        team.printAll();
49
50        team.updateSalary(firstname: "Alice", salary: 3500);
51
52        team.raiseAll();
53
54        team.printAll();
55    }

```

Problems Output Debug Console **Terminal** Ports

```

PS D:\Lab1> ^C
PS D:\Lab1>
PS D:\Lab1> d;; cd "d:\Lab1"; & "C:\Program Files\Microsoft\jdk-21.0.12.8-hotspot\bin\java.exe" "-XX:+ShowCodeDetailsInExceptionMes
Boss:
John
Smith
5000.0
Employee:
Alice
Brown
3000.0
Boss:

```

```

J q4.java > java > q4 > balance
1 public class q4 {
2
3     private String name;
4     private double balance;
5
6     public q4(String name, double balance) {
7         this.name = name;
8
9         if (balance > 0) {
10             this.balance = balance;
11         } else {
12             this.balance = 0;
13         }
14     }
15
16     public void deposit(double depositAmount) {
17         if (depositAmount > 0) {
18             balance += depositAmount;
19         }
20     }
21
22     public String getName() {
23         return name;
24     }
25
26     public void setName(String name) {
27         this.name = name;
28     }
29
30     public double getBalance() {
31         return balance;
32     }
33
34     Run main | Debug main | Run | Debug
35     public static void main(String[] args) {
36
37         q4 account = new q4(name: "John", balance: 100);
38
39         System.out.println(account.getName());
40         System.out.println(account.getBalance());
41
42         account.deposit(depositAmount: 50);
43
44         System.out.println(account.getBalance());
45     }

```

Problems Output Debug Console Terminal Ports

```

PS D:\Lab1>
PS D:\Lab1> d; cd 'd:\Lab1'; & 'C:\Program Files\Microsoft\jdk-21.0.12.8-hotspot\bin\java.exe' -cp . q4
John
100.0
150.0
PS D:\Lab1> ^C
PS D:\Lab1>
PS D:\Lab1> d; cd 'd:\Lab1'; & 'C:\Program Files\Microsoft\jdk-21.0.12.8-hotspot\bin\java.exe' -cp . q4
John
100.0
150.0
PS D:\Lab1>

```

```
q5.java > java > ^q5 > main(String[] args)
1 public class q5 {
2
3     private q4[] accounts;
4
5     public q5(int maxSize) {
6         accounts = new q4[maxSize];
7     }
8
9     public boolean appendAccount(q4 account) {
10         for (int i = 0; i < accounts.length; i++) {
11             if (accounts[i] == null) {
12                 accounts[i] = account;
13                 return true;
14             }
15         }
16
17         return false;
18     }
19
20     public q4 getAccount(int index) {
21         if (index >= 0 && index < accounts.length && accounts[index] != null) {
22             return accounts[index];
23         }
24
25         System.out.println(x: "Invalid account index");
26         return null;
27     }
28
29     Run main | Debug main | Run | Debug
    public static void main(String[] args) {
30
31         q5 list = new q5(maxSize: 2);
32
33         q4 account1 = new q4(name: "John", balance: 100);
34         q4 account2 = new q4(name: "Alice", balance: 200);
35
36         System.out.println(list.appendAccount(account1));
37         System.out.println(list.appendAccount(account2));
38
39         System.out.println(list.getAccount(index: 0).getName());
40         System.out.println(list.getAccount(index: 0).getBalance());
41     }
42 }
```

Problems Output Debug Console Terminal Ports

```
PS D:\Lab1> ^C
PS D:\Lab1>
PS D:\Lab1> d:; cd 'd:\Lab1'; & 'C:\Program Files\Microsoft\jdk-21.0.12.8-hotspot\bin\java.exe' '-XX:
true
true
John
100.0
PS D:\Lab1>
```

```

1
2 public class Employee {
3
4     private String firstname;
5     private String lastname;
6     private double monthllysalary;
7
8     public Employee(String firstname, String lastname, double monthllysalary) {
9         this.firstname = firstname;
10        this.lastname = lastname;
11        this.monthllysalary = monthllysalary;
12    }
13
14    public Employee(String firstname, String lastname) {
15        this.firstname = firstname;
16        this.lastname = lastname;
17        this.monthllysalary = 0;
18    }
19
20    public String getFirstname() {
21        return firstname;
22    }
23
24    public String getLastName() {
25        return lastname;
26    }
27
28    public double getMonthllysalary() {
29        return monthllysalary;
30    }
31
32    public void setFirstname(String firstname) {
33        this.firstname = firstname;
34    }
35
36    public void setLastName(String lastname) {
37        this.lastname = lastname;
38    }
39
40    public void setMonthllysalary(double monthllysalary) {
41        if (monthllysalary > 0) {
42            this.monthllysalary = monthllysalary;
43        }
44    }
45
46    Run main | Debug main | Run | Debug
47    public static void main(String[] args) {
48
49        Employee employee1 = new Employee(firstname: "John", lastname: "Smith", monthllysalary: 3000);
50        Employee employee2 = new Employee(firstname: "Alice", lastname: "Brown");
51
52        System.out.println(employee1.getFirstname());
53        System.out.println(employee1.getLastName());
54        System.out.println(employee1.getMonthllysalary());
55
56        System.out.println(employee2.getFirstname());
57        System.out.println(employee2.getLastName());
58        System.out.println(employee2.getMonthllysalary());
59
60        System.out.println("Yearly salary: " + employee1.getMonthllysalary() * 12);
61
62        employee1.setMonthllysalary(employee1.getMonthllysalary() * 1.10);
63
64        System.out.println("After 10% raise: " + employee1.getMonthllysalary());
65    }
66 }

```

blems Output Debug Console Terminal Ports

```

PS D:\Lab1> ^C
PS D:\Lab1>
PS D:\Lab1> d; cd 'd:\Lab1'; & 'C:\Program Files\Microsoft\jdk-21.0.12.8-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionStack'
John
Smith
3000.0
Alice
Brown
0.0
Yearly salary: 36000.0
After 10% raise: 3300.0000000000005
PS D:\Lab1>

```